

Color and Shade Matching

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There is a significant professional interest in color and appearance in dentistry.[1,2] The number of publications listed on MEDLINE for the keywords *color* and *dentistry* has increased rapidly from only 109 in 1970 to 4,315 in 2010. The same is true for the public interest and demand related to esthetic and cosmetic dental procedures. According to a survey by the American Academy of Cosmetic Dentistry, virtually everyone believes that a smile is an important social asset, and three-quarters of adults feel that an unattractive smile can hurt a person's chances for career success.[3] Discolored, yellow, or stained teeth were identified as characteristics that make a smile unattractive. On the other hand, participants listed straightness, whiteness, and color of teeth as the first things they notice in a person's smile. The top two items related to desired changes in one's smile were whiter and brighter teeth. It is easy to notice that almost all of these remarks are associated with tooth color.

Therefore, color is a key component of the esthetic complex in dentistry, and to a significant extent color delineates *esthetic* from *nonesthetic* in our profession. A simple explanation on the importance of color in dentistry was offered by Dr Stephan Bergen, one of the pioneers of esthetic dentistry: "Color is unimportant to the physiologic success of a dental restoration, yet it could be the controlling factor

in the overall acceptance by the patient."[4]

Color

Color is a psychophysical sensation produced in the eye by visible light and interpreted by the brain. Color requires three components: light source, object, and observer.

A *light source* is any area or body that emits radiation in the visible spectral range. Some standard light sources are designated as A (incandescent), B, C, and D (daylight). Daylight is further categorized as D50, D55, D65, and D75, the designations corresponding to the correlated color temperature (CCT) in kelvin (the number after the letter *D* times 100 equals the CCT in kelvin). The *object* reflects, absorbs, or transmits the light from the source.[5] The reflected portion of light is used during shade matching in dentistry. The *observer* is the third part of the triplet. All visual sensations are brought to the brain through the eye. The retina has two types of nerve endings: rods and cones. While rods only record light (ie, they respond to black and white), cones enable color vision. The three types of cones are blue-, green-, and red-sensitive, or short, medium, and long wavelengths, respectively. Visual information from the eye is then relayed to the brain, where the messages from the rods and cones are interpreted.

Dimensions of color

Color is a three-dimensional phenomenon characterized by the dimensions of hue, value (lightness), and chroma (Fig 4-1).

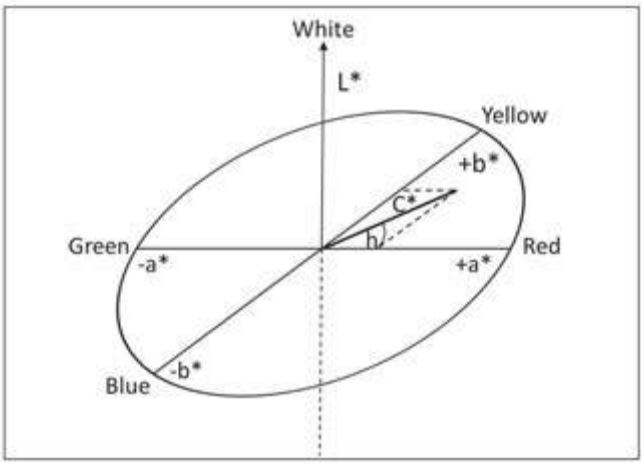


Fig 4-1 Color coordinates L*a*b*C*h. L*—lightness (vertical axis); a* (–a*, green; +a*, red); b* (–b*, blue; +b*, yellow); C^*_{ab}—chroma; h_{ab}—hue.

Hue is a dimension that enables differentiation between "color families." For example, red, green, blue, and yellow represent hues (Fig 4-2). Graphically, hues are placed on the exterior of a closed hue circle that encompasses 360 degrees (see Fig 4-1). *Value,* or lightness, enables differentiation between light and dark colors (Fig 4-3). In the center of the hue circle exists a vertical value axis that progresses from pure black at the bottom to pure white at the top (see Fig 4-1). *Chroma* allows differentiation between pale versus strong, pale versus chromatic, and washed-out versus saturated colors (Fig 4-4). Chroma is represented as the distance from the achromatic value axis toward the particular hue on the circumference of the circle, and it increases with the increase of the distance from the vertical axis (see Fig 4-1). While hue is related to different colors, chroma is associated with the differences in strength of the same color (eg, pale red versus chromatic red).



Fig 4-2 Hue. Color families or color name.



Fig 4-3 Value. White to black, achromatic.



Fig 4-4 Chroma. Pale to strong, washed-out to chromatic.

Color notation systems

Various color notation systems have been used in dentistry. Historically, these include the Munsell hue-value-chroma (HVC) system and two formulas developed by the International Commission on Illumination (Commission Intérnationale de l'Eclairage [CIE]), CIE 1931 (xyY) and CIE 1964 (XYZ).

However, the most frequently used color notation system in dentistry is the CIELAB (CIE L*a*b* or CIE 1976) system[5] (see Fig 4-1). Any color can be designated by one of two classifications of coordinates related to the diagram in Fig 4-1. Rectangular coordinates of the CIELAB color space are L* (lightness), a* (–a*, green; +a*, red), and b* (–b*, blue; +b*, yellow), which mark a point on the vertical (L*) axis and on the circumference of the horizontal wheel. Polar coordinates are L*,

C^*_{ab} (chroma) and h_{ab} (hue), which mark a point on the vertical (L*) axis as well as the distance away from the vertical axis (C^*_{ab}) at a particular angle (h_{ab}) on the horizontal plane.

CIELAB color difference is designated as ΔE^*_{ab}, where the symbol Δ stands for difference and E stands for *Empfindung*, which means *sensation* in German. CIELAB color difference takes into account all aspects of color and is calculated using differences in either L*a*b* or light, chroma, and hue coordinates.

More recently, an advanced CIE color-difference formula, CIEDE2000, has been introduced.[6,7] Color difference in CIEDE2000 is designated as $\Delta E'$. The CIEDE2000 formula is becoming more common in dental research as well.[8]

Additional Appearance Attributes

Tooth appearance attributes other than color include opacity, translucency, gloss, opalescence, fluorescence, and iridescence.

Opacity and translucency are exactly of opposite nature. *Opacity* is the ability of a material to block the passage of light through the material, whereas *translucency* is the ability of the material to allow light to pass through it.[2,9]

Gloss has physical, physiologic, and psychologic aspects. The following visual criteria define an object's gloss: specular gloss, contrast gloss (or luster), sheen, distinctness of image, haze, and surface phenomena. *Specular gloss* is the most frequently evaluated type in dentistry, and it corresponds in simple terms to the ratio of reflected to incident light energy, where the angle of the reflected light is equal and opposite to the angle of the incident light for a specified angle of incidence, source, and receptor angular apertures.[2,9]

Opalescence is an optical property that gives a material a bluish appearance under reflected light and an orange/brown appearance under transmitted light, while *fluorescence* is the absorption of light by a material and the spontaneous emission of light in a longer wavelength. Fluorescence is not usually detected with the color-corrected lights normally used in dental operatories; it is normally seen under ultraviolet lighting (eg, "black light" or sunlight). It is important for optimal esthetic matching that dental materials, such as porcelains, especially when used in veneers on the anterior teeth, have fluorescing agents in their composition; otherwise they may appear rather dull and not lifelike. *Iridescence* is the rainbowlike effect in which surfaces appear to be different colors depending on the illuminating and/or

viewing angle.[2,9]

Color and Appearance of Human Teeth

Tooth color varies from person to person, by dentition (primary versus permanent), by tooth within the same dentition, and within the same tooth. Permanent teeth are small, curved, whitish-yellowish, and slightly reddish. They exhibit color transitions from cervical to incisal, mesial to distal, and labial/buccal to lingual. Variations in thickness and translucency of the enamel and dentin are also present. In addition, tooth color changes throughout a person's lifetime due to various physiologic and pathologic conditions. Local color characteristics, such as enamel cracks and craze lines, enamel hypoplasia, fluorosis, tetracycline staining, or incisal halo, bring additional complexity to tooth color matching, communication, reproduction, and verification.[10,11] Interaction among color-related characteristics of human teeth and other attributes of appearance, such as translucency/opacity, gloss, fluorescence, and phosphorescence, makes working with color one of the most challenging tasks in esthetic and restorative dentistry (Fig 4-5).

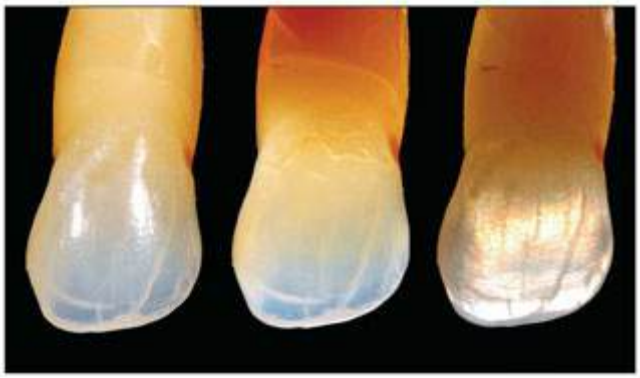


Fig 4-5 Interaction among color-related characteristics of human teeth and other attributes of appearance makes working with color one of the most challenging tasks in dentistry. (Courtesy of Adam J. Mieleszko, CDT, New York, New York.)

There are findings on the influence of sex, bleaching history, and smoking on the color of permanent teeth. A color difference (ΔE^*) of 3.0 was recorded between teeth in males and females. Male teeth were significantly darker, more chromatic, and redder than female teeth. The color difference between bleached and nonbleached teeth was even greater ($\Delta E^* = 4.6$); nonbleached teeth were significantly darker, more chromatic, and of lower hue angle. As for the influence of smoking on tooth color, the color difference between nonsmokers and smokers was 2.7, with only the difference in lightness being statistically significant.[12] A

significant color difference (ΔE* = 8.2) has been shown between permanent and primary teeth, and primary teeth were found to be much lighter (ΔL* = 7.7), less chromatic (ΔC* = –2.5), and redder (Δh° = –4.2) than permanent teeth.[13]

Color-Matching Ability

Influence of sex and experience

The shade-matching abilities of color-normal men and women are equal. Therefore, sex should not be the criterion for choosing the person to perform shade matching. Shade-matching competency of dental students and professionals should be routinely tested.[14] Color deficiency is a weakness or absence in one or more cone systems. People with deficiencies in their cone systems are not "color blind"; rather, approximately 8% of males and 0.5% of females are color deficient. Different types of color deficiency compared with color-normal vision are illustrated in Fig 4-6. The literature is equivocal on whether years in practice or specialty affects shade-matching ability. Experience is relevant, but so is the status and age of the visual organ.[9]

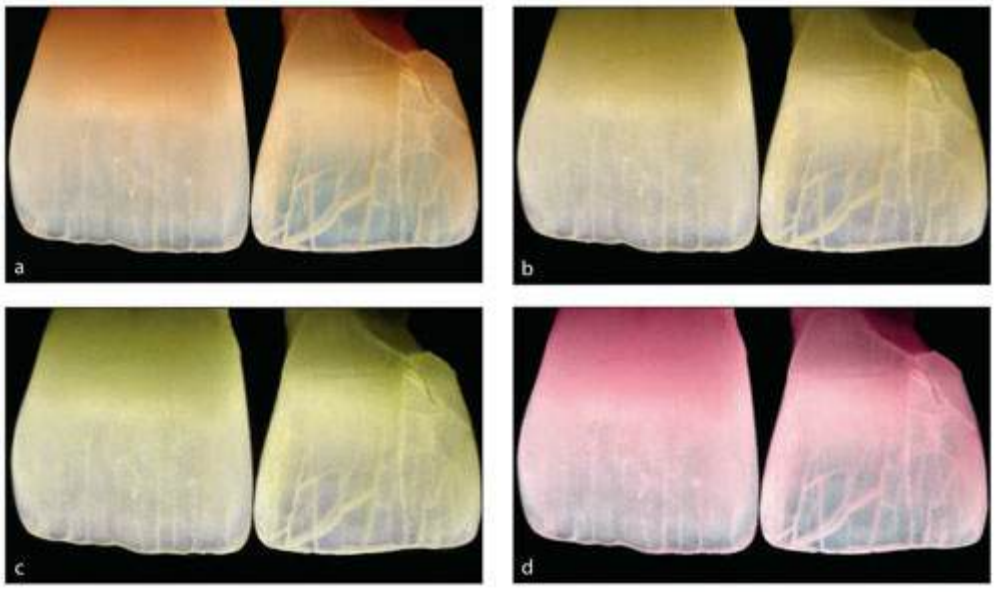


Fig 4-6*(a)* Image representing normal color vision. (Courtesy of Adam J. Mieleszko, CDT, New York, New York.) *(b to d)* Vischeck computer simulations representing different types of color deficiency: deuteranope (green photoreceptors of retina are absent) *(b)*, protanope (complete absence of red photoreceptors) *(c)*, and tritanope (blue receptors are absent) *(d)*.

Color education and training

Sproull stated in 1973 that "the technology of color is not a simple matter that can be learned without study; neither is it a complicated matter beyond the comprehension of dentists."[15] Color education and training can influence shade-matching results. Education and training should be the first step in the process, which would ideally result in predictable and enhanced esthetic outcomes.[1]

It is apparent that humans are not as good in color matching and reproduction as they might expect. When asked to match 16 pairs of tabs of two VITA Classical shade guides (Vident) using a visual method, the participants in a study correctly matched only 50%.[16] In another study, which more closely resembled clinical dentistry because no exact match was included in the choices, the observers' choice was the second or the third best match.[17]

Several color education and training programs are currently available.[1,10,11] Dental Color Matcher,[10] a free online color education and training program for esthetic dentistry, is in essence a computer game. Dental Color Matcher consists of interactive color-matching exercises using the VITA Linearguide 3D-Master shade guide (Vident) ("closest match," "exact match," and "match the pairs" exercises [Fig 4-7]), a didactic video, and a quiz. Upon program completion, all users can print a diploma and obtain two continuing education credit hours.

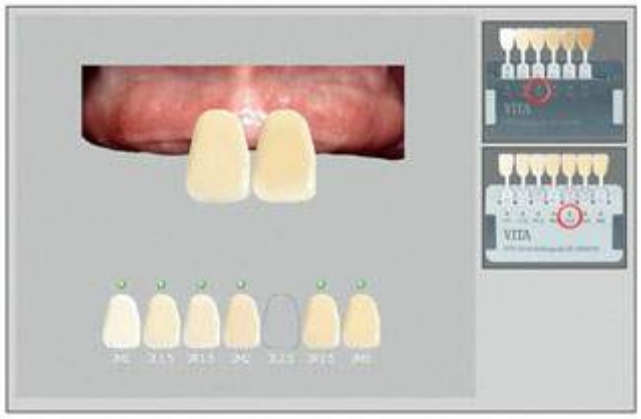


Fig 4-7 Dental Color Matcher: "closest match" exercise.

Color Matching and Measuring

Tooth color can be matched or measured visually and/or using color-measuring instruments. Dental shade guides are used for making visual comparisons, and this shade-matching method is predominant in dentistry today,[18] despite the fact that it is subjective and to a certain extent inconsistent. Dental color-measuring devices are objective, and while very beneficial, they are presently used by a relatively small

percentage of dental professionals. There is a need to carefully consider the meaning of the words *subjective* and *objective* when it comes to color in dentistry. While there is a difference in subjective color perception among color-normal individuals as well as for the same individual, these differences are typically not very pronounced. On the other hand, errors in instrument selection and use, differences from visual findings, and instrument inability to quantify optical illusions are some of the shortcomings associated with objective color matching. One should not forget that the design of all advanced color-measuring instruments incorporates the response of the so-called standard observer. Therefore, the objective might be good only as far as it corresponds to the subjective.

Dental shade guides

The primary requirements for dental shade guides are logical ordering and adequate color distribution within the tooth color space.[19–22] Three basic conceptions of dental shade guides are currently present:

1. VITA Classical and Classical-keyed products (empirical conception)
2. VITA 3D-Master shade guides (scientifically grounded)
3. Other (proprietary conceptions)

Several VITA Classical-keyed and proprietary shade guides, and the combinations of these two, are shown in Fig 4-8.

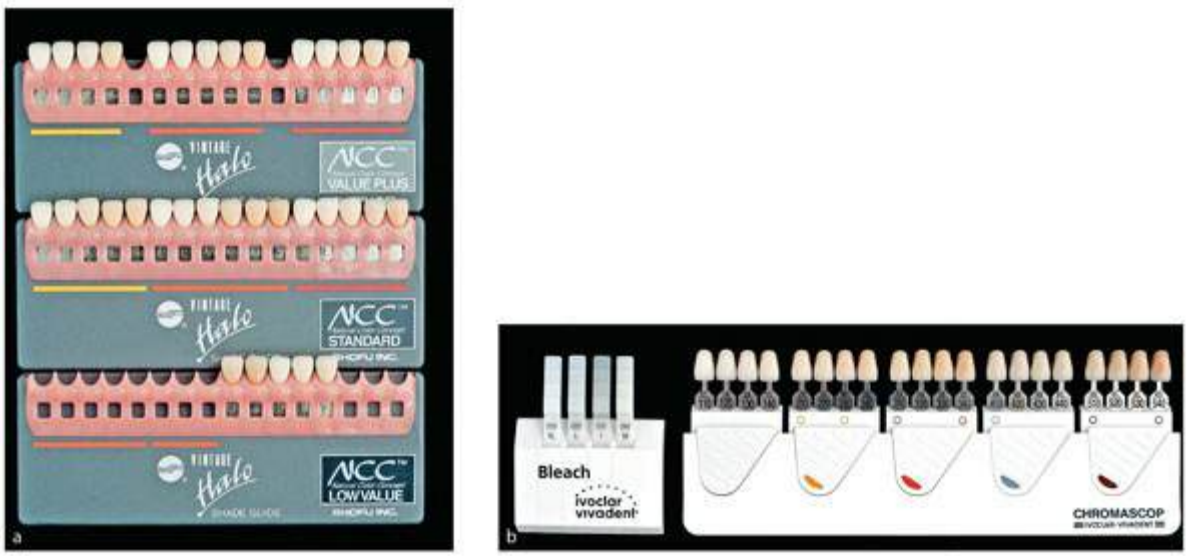


Fig 4-8 Examples of VITA Classical-keyed and proprietary shade guides and combinations of these two: *(a)* Vintage Halo NCC (Shofu), *(b)* Ivoclar Chromascop (Ivoclar Vivadent), *(c)* Venus (Heraeus) *(top)* and Estelite (Tokuyama) *(bottom)*, and *(d)* Esthet-X (Dentsply/Caulk) *(top)* and

Trubyte Bioform (Dentsply/Caulk) *(bottom)*.

VITA Classical

The VITA Classical shade guide has been available since 1956. Since that time, it has been the gold standard for visual shade matching. The majority of resin composites are keyed to the VITA Classical shade guide system. Two tab arrangements are utilized for this system.

The VITA Classical *A-to-D arrangement* (Fig 4-9) distributes tabs into four groups based on hue: *A* is red, *B* is yellow, *C* is gray, and *D* is reddish-gray. Within each group, chroma increases with an increase in the tab number, which appears after the letter designating the group (eg, B4 is more saturated than B1). The shade-matching method with the tabs arranged from A to D encompasses initial selection of the group in the first step, followed by selection of a shade within the chosen group in the second step.

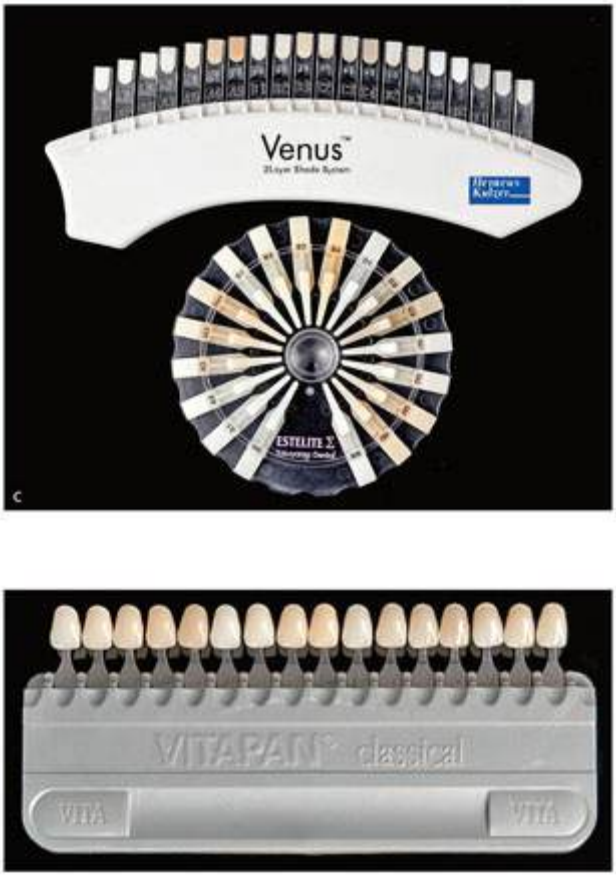


Fig 4-9 VITA Classical shade guide, A-to-D arrangement. From left to right: A1, A2, A3, A3.5, A4, B1, B2, B3, B4, C1, C2, C3, C4, D1, D2, D3, D4.

The VITA Classical *value scale arrangement* (Fig 4-10) represents a light-to-

dark arrangement, from 1 (B1) to 16 (C4). The only shade-matching instruction for this arrangement of tabs is to select the best overall match. The value scale is frequently used for visual monitoring of tooth-whitening efficacy, expressed in shade guide units (SGU). However, discrepancies in the visual light-to-dark arrangement and nonuniform color distribution in the VITA Classical value scale reduce to a certain extent its efficacy for this purpose.



Fig 4-10 VITA Classical shade guide, value scale arrangement. From left to right: B1, A1, B2, D2, A2, C1, C2, D4, A3, D3, B3, A3.5, B4, C3, A4, C4.

VITA 3D-Master

VITA 3D-Master shade guides are the Toothguide, Linearguide, and Bleachedguide. These shade-matching tools have been developed based on research findings.[16,23–26] They match the color range and distribution of human teeth better than other shade guides, so the probability of selecting a good match will be highest with these products.[27–29] Each 3D-Master tab is marked using a number-letter-number combination. The first number designates the group and represents value, with 0 as the lightest and 5 the darkest. The letter represents hue, with the letter *L* corresponding to yellowish, *M* to medium, and *R* to reddish. The final number

represents chroma, with 1 being the least chromatic and 3 the most chromatic.

Toothguide was the first 3D-Master shade guide. Shade matching with the Toothguide requires three steps. In the first step (Fig 4-11), the lightness is determined using all tabs, and the number of possible shades is reduced from a total of 29 to 2 (group 1), 3 (groups 0 and 5), or 7 (groups 2, 3, and 4). Vertical chroma variations are present within each group (Fig 4-12), while horizontal hue variations exist only within groups 2, 3, and 4 (Fig 4-13). Chroma and hue should be determined in two separate subsequent steps. This shade-matching method can be challenging for those with little experience in tooth shade matching or with no or little knowledge of color physics.

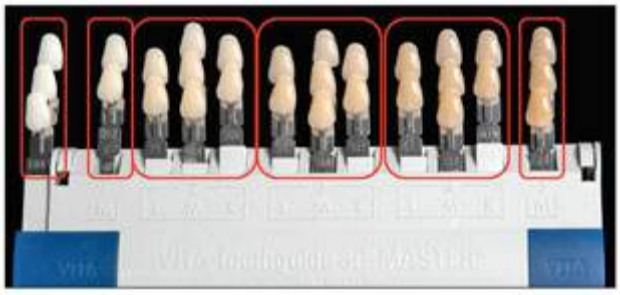


Fig 4-11 VITA Toothguide 3D-Master: Group division (0 to 5) is based on lightness (value).

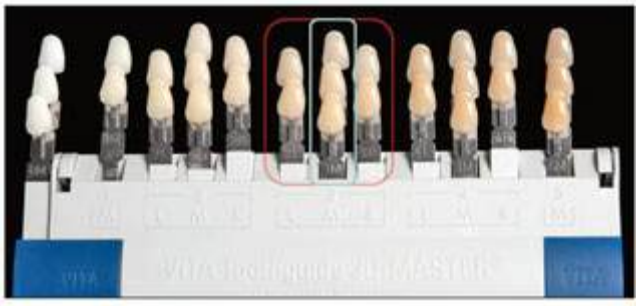


Fig 4-12 VITA Toothguide 3D-Master: Vertical chroma variations (ranging from 1 to 3) are present within each group.

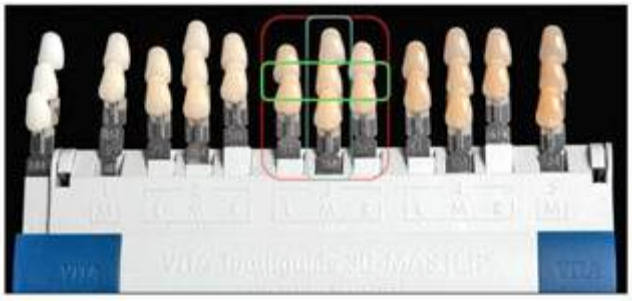


Fig 4-13 VITA Toothguide 3D-Master: Horizontal hue variations (L, M, and R) are present only within groups 2, 3, and 4.

Linearguide 3D-Master has the same tabs as the Toothguide. However, its design and recommended shade-matching method are different. A single linear scale consisting only of the middle tabs from each group (0M2 to 5M2, dark-gray tab holder) is used in step 1 (group selection). A small number of tabs (six) with pronounced color differences and the familiar linear tab arrangement simplify the initial selection (Fig 4-14a). The second step involves "fine tuning" within the group selected in step 1: 0-1, 2, 3, 4, or 5 (Fig 4-14b, light-gray holders). Users frequently describe the shade-matching method with Linearguide as self-explanatory and user-friendly.[17]

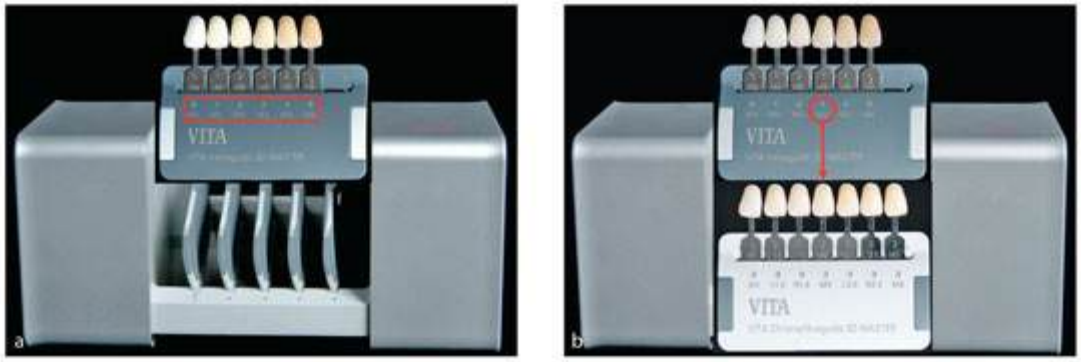


Fig 4-14 VITA Linearguide 3D-Master. *(a)* Step 1: group selection. *(b)* Step 2: selection within the group.

The 29-SGU Bleachedguide 3D-Master is the first shade guide developed for visual evaluation of tooth whitening (Fig 4-15). It exhibits a wider color range and a more consistent color distribution when compared with the VITA Classical shade guide and some other products. The inclusion of very light shades complements contemporary esthetic dentistry.[21,22] Bleachguide 3D-Master has recently replaced VITA Classical as an example of an appropriate shade guide for the evaluation of tooth whitening in the American Dental Association guidelines for bleaching products (dentist dispensed and over-the-counter) as well as for stain-removal products.[30]

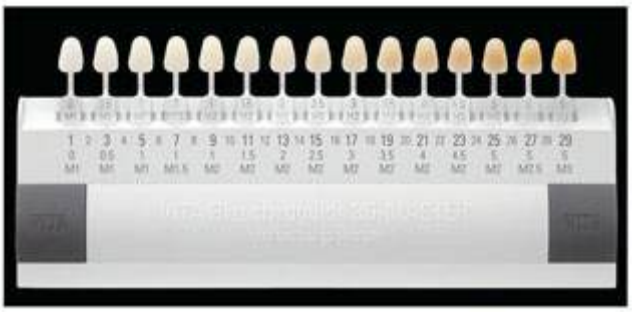


Fig 4-15 Prototype of a 29-step VITA Bleachedguide 3D-Master.

Shade-matching conditions

Determining the optimal shade-matching conditions requires a consideration of light and environment. Either a ceiling light or a portable light (floor, table, or handheld light), can be used for shade matching. In either case, diffuse color-corrected light should be used for color matching and reproduction in the office and laboratory. Daylight (D65, D55, or similar) lamps and tubes with a color rendering index (CRI) of 90 or greater should be selected. The recommended light intensity is 1,000 to 1,500 lux. The immediate surroundings should ideally be neutral, such as light gray. Creation of a designated shade-matching and reproduction area in the office/laboratory is helpful.

Using handheld lamps (Fig 4-16) for shade matching can significantly reduce the influence of surrounding colors, especially if the ceiling light is off. These lamps usually come with instructions on the positioning, distance, and specific method to be used.



Fig 4-16 Handheld lamps for shade matching. From left to right: Optilume Trueshade (Optident), shade wand (Authentic Dental Lab), Rite-lite (AdDent), and Demetron Shade Light (Kerr).

Guidelines for shade matching

- Shade matching should be performed by individuals with normal color vision; color-deficient persons should be assisted. Tinted glasses or contact lenses should not be worn during shade matching or any other work with color in dentistry.
- The tooth whose color is to be matched should be cleaned with pumice or toothpaste and rinsed. The patient should remove any lipstick.
- To avoid a mistake in shade matching due to eye fatigue of the dentist, the procedure should be performed at the beginning of the appointment.
- The viewing angle should be approximately 45 degrees to the illumination angle. In the dental operatory, the tooth should be observed perpendicularly to its labial surface, with the eyes aligned at the same level as the teeth. Shade-matching

distance should be 25 to 35 cm (or 10 to 14 inches). This corresponds to the viewing angle of 0 degrees, while the light (one-directional, two-directional, or circumferential) should be at a 45-degree angle to the tooth. In the dental laboratory, it is more typical that a table light is placed perpendicular to the dentition (a 0-degree lighting angle). In this case, the teeth should be viewed from approximately a 45-degree viewing angle.

- Whenever possible, the shade tabs should be placed on the same plane with the tooth to be matched and with the same relative edge position. When the adjacent tooth is present, tabs can be placed in front of it or horizontally or vertically in between the maxillary and mandibular teeth.
- A single shade-matching trial should last no more than 5 seconds at a time, because a vision pigment is used up quickly during color perception. A neutral, light-gray card should be observed between trials. In the past, light-blue cards were recommended, but they are now discouraged because of the after-effect phenomenon: When the blue cones are fatigued, a neutral field may appear slightly yellow.
- Wetting with water can reduce or neutralize surface texture differences between the tab and the tooth.
- The number of potential matches should be reduced as quickly as possible. These tabs should be separated from the other tabs before proceeding with final shade matching.
- Translucency, gloss, and local color characteristics should be visualized and documented.
- It is always good to verify the shade selection under different lights and at different angles to avoid or reduce metamerism. *Metamerism* occurs when two colored samples appear to match under one set of observational conditions but not under another.

The human eye cannot see lightness, chroma, and hue separately. Therefore, color matching is a comparison of the overall impression of color differences between observed objects. Humans can see differences in lightness, chroma, and hue, but these differences are dependent on one another. It is quite easy, for example, to confuse a decrease in chroma with an increase in lightness. The example with which many dental professionals are familiar is the A1 and B1 tabs of VITA Classical (B1 is considered to be the lightest shade in the system). Although the A1 tab actually has a higher lightness (value) than B1, it appears darker because it is more chromatic. Therefore, the best shade-matching advice and strategy is to simply select the best

match. When one must perform a dimension-by-dimension technique, identify value first, then chroma, and then hue.

Color-matching instruments and software

Color-matching instruments and software can be used independently or combined with a visual method. They can also be valuable for color communication, reproduction, and verification. Spectrophotometers and colorimeters are available, and some of them are capable of providing a detailed image of the tooth surface as well as a tooth color map.[31] An overview of available instruments and software is listed in Table 4-1.

Table 4-1 **Instruments and software for color matching in dentistry**[9,10,32]

	ClearMatch	Crystaleye	Easyshade Compact	SpectroShade Micro
Manufacturer	Clarity Dental	Olympus	Vident	MHT Optic Research
Device type	Software	Spectrophotometer	Spectrophotometer	Spectrophotometer
Measurement area	Complete tooth image	Complete tooth image	5-mm probe diameter	Complete tooth image
Relative cost	Low	High	Low	Moderate

Shade-matching results of instrumental measurements are typically given in the units of the CIELAB system and the best-matching shade tab(s). All listed compact and cordless color-matching instruments consist of a handpiece and a base unit (Figs 4-17 and 4-18). They have their own light sources and a means for electronic storage and transfer of information.[10] Tooth color–matching instruments can be categorized into complete tooth surface measurement devices and limited-area ("spot") measurement devices.



Fig 4-17 Shade-matching instrument: Spectro-shade Micro (MHT Optic Research).



Fig 4-18 Shade-matching instruments: Vita EasyShade Compact (Vident) *(left)* and Crystaleye (Olympus) *(right)*.

SpectroShade Micro (MHT Optic Research) uses a digital camera/light-emitting diode (LED) spectrophotometer combination with an internal computer, analytic software, and a tooth-positioning guidance system[31] (see Fig 4-18). Crystaleye (Olympus) uses a combination of a traditional spectrophotometer and digital photography (see Fig 4-18). Virtual shade tabs in its database can be cross-referenced and superimposed with a natural tooth or dental restoration, thus enhancing color communication.[33] Vita Easyshade Compact (Vident) is a contact-type spectrophotometer with a 5-mm probe diameter and a variety of measurement modes (see Fig 4-18): tooth single mode, tooth area mode (cervical, middle, and incisal shades), restoration color verification (includes lightness, chroma, and hue comparison), and shade tab mode (practice/ training mode).[32,34] ClearMatch (Clarity Dental) is software that requires the use of a digital camera. It compares tooth color

with its own color database.

Instrumental shade matching has attracted a lot of attention from researchers,[10] starting from the performance assessment of shade-matching devices[35–37] and digital imaging systems[38–40] through comparisons between visual and instrumental findings,[41–43] evaluation of color of natural teeth[27,28,44,45] and various dental materials,[46–49] and tooth whitening.[50–53] While many of these systems have been shown to be reproducible and accurate and can be very helpful in providing quantitative differences, they will likely always be used as adjuncts to human shade matching because of the highly subjective and somewhat personal nature of the process.

Communicating color and appearance

Once the shade for indirect restorations is selected, analyzed, and agreed upon by the patient, the information should be transferred to the dental laboratory. In addition to the information obtained using visual and instrumental shade analysis, subsequent reference photography of the tooth, with the selected shade tab aligned next to it, is highly recommended as an effective communication tool (Fig 4-19). Uniformly calibrated monitors in the dental office and laboratory further contribute to enhanced communication on color and appearance. Digital imaging can be shared instantly via the Internet, email, CD-ROM, or other storage device. Inclusion of verbal and written instructions and custom-made or modified shade tabs is also recommended for complex cases.



Fig 4-19 Communication on color and appearance using various shade-matching techniques and digital photography. *(a)* Photographs should be taken so the shade tab designation is included in the photograph and can be viewed by the laboratory. *(b)* The black-and-white tab attached to the shade tab handle is used by the dental laboratory to calibrate the color of their computer monitor to the photograph.

Reproduction and Verification of Color and

Appearance

Color reproduction is a combination of dental art and science. Because different restorative techniques are described in other parts of this book, the three clinical cases presented here include only information about color-related properties of dental materials and offer examples of nonconventional yet very successful techniques for reproducing color and appearance.

Color-related properties of dental materials

Color-related properties of esthetic dental materials can be categorized as follows[1,9]:

- Color compatibility between materials and teeth, between different materials, and within the same material (batch color variations)
- Color stability during fabrication (condensation, firing, glazing, or at placement), polymerization and other types of setting, and after placement (aging and staining)
- Color interactions (blending and layering)

It is important to note that in spite of remarkable improvements associated with the optical properties of dental materials, one person can still experience different outcomes based solely on material selection.[9] Therefore, material selection can be critical for the esthetics and, consequently, for the overall success of a dental restoration. It is highly recommended to consult professional resources, such as MEDLINE, professional journals and newsletters,[53–59] and information from manufacturers regarding material selection. The following cases highlight different techniques for achieving excellent esthetic outcomes.

Case A: Five dimensions of color

The five dimensions of color technique (Fig 4-20) was developed by Lorenzo Vanini primarily for restoring natural color and appearance of teeth with direct resin composite restorations.[60,61] This approach encompasses the following elements:



Fig 4-20 Five dimensions of color technique. Facial *(a)* and palatal *(b)* views of the clinical situation show a nonvital, discolored, and severely compromised maxillary left central incisor with a large existing composite restoration. Following root canal retreatment and use of the internal walking-bleach technique with 12% hydrogen peroxide, the tooth is prepared with a chamfer on the facial enamel *(c)* and 90-degree butt interproximal and lingual margins *(d)*. *(e)* A silicone index made from a mock-up or diagnostic wax-up is used for creation of the lingual wall and is adapted to all involved teeth. *(f)* A thin layer of universal enamel resin composite allows for easy reproduction of the natural anatomy of the lingual wall. Dentin body and enamel effects are then created: *(g)* dentin body is formed with lighter incisal mamelons (characterization type 1); *(h)* amber and white resin composite are used at the incisal and approximal margins (characterization type 3); *(i)* free spaces between margin and dentin mamelons are filled with opalescent blue (opalescent type 1) to increase the halo effect; *(j)* a thin layer of intense white is stratified to create a white-spot appearance (intensive type 2). The stratification is completed on the facial surface with a thin layer (0.8 mm) of universal enamel. In the facial *(k)* and palatal *(l)* views, the intensive, opalescent, and characterization elements give a natural, three-dimensional quality to the completed case. (Courtesy of Lorenzo Vanini, Chiasso, Switzerland.)

- Chromaticity is related to one hue (universal dentin) and different chroma levels (0, 0.5, 1, 2, 3, 4, 5, and 6).

- Value (luminosity) is strictly related to enamel, with two main groups: lower value (older tooth biotype) and higher value (younger tooth biotype).
- Intensives are classified by shape types: spots, small clouds, snowflakes, and horizontal bands.
- Opalescents create the blue and amber hue of the incisal halo at the interproximal level and free enamel margin. The author describes different shapes: mamelon, split mamelon, comb-like, windowlike, and stainlike.
- Characterizations encompass five different types: two in dentin, mamelon, and band, and three in enamel, margin (younger teeth), stain, and cracks (adult and older teeth).

Case B: Natural layering concept

The natural layering concept (Fig 4-21) was developed by Didier Dietschi for the creation of highly esthetic direct restorations.[62,63] This method suggests single hue, single opacity, and broad chroma scales for mimicking dentin color and appearance. Enamel and corresponding materials are categorized as the following types: young enamel (white tint, high opalescence, less translucency), adult enamel (neutral tint, less opalescence, and intermediary translucency), and old enamel (yellow tint, more translucency). Only two basic layers, dentin and enamel, are used to enhance tooth and smile anatomy.



Fig 4-21 Natural layering concept. *(a)* A young patient presents with a healthy dentition but a highly irregular smile line. The patient wants improvement but demands a conservative solution. *(b)* Following placement of rubber dam, an index (made from a mock-up or wax-up) will serve to

guide proper placement of the two resin composite layers in three dimensions. *(c)* The restorations are placed using the natural layering concept. *(d)* The completed restorations demonstrate the satisfactory esthetic integration of the direct composite restorations using a modern hybrid material, which exhibits proper color, opacity, translucency, and opalescence. (Courtesy of Didier Dietschi, Geneva, Switzerland.)

Case C: Combined visual-instrumental technique

Shigemi Nagai and her team are among the leaders when it comes to combining visual and instrumental techniques[64–67] (Fig 4-22). Using sophisticated color-measurement instruments to their best advantage requires clinical/laboratory excellence and sound color science. This means not only knowing how, when, and why to use the instrument but also being able to understand and interpret the findings and to adjust the color and appearance of the dental restoration using principles of subtractive color mixing and complementary color theory, for both color control and final verification of the crowns and gingiva. This case provides an example of what is possible when this combination technique is utilized by experts in the field.



Fig 4-22 Combined visual-instrumental technique. A 32-year-old woman presents with missing maxillary lateral incisors. She wants to replace them with single implants and crowns. *(a to d)* Two narrow-diameter bone-level implants are placed, and zirconia all-ceramic crowns (Lava, 3M ESPE) are fabricated with Vita VM9 layered porcelain (Vident). *(e)* Although the color of the crowns is symmetric, a perceivable difference in the peri-implant soft tissue appearance is observed between the two implant-supported crowns. *(f and g)* Color mapping of the peri-implant tissue around the maxillary right lateral incisor shows that it has blanched, which can also be seen in the linked image. *(h)* At the 1-month follow-up appointment, an improvement in the optical properties of the peri-implant soft tissue of the maxillary right lateral incisor compared with *f* is observed. *(i)* The digital camera image shows an excellent color match of both maxillary lateral incisor crowns and an improved appearance of the peri-implant soft tissue. (Courtesy of Shigemi Nagai, Boston, Massachusetts.)

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